Probability — Experimental probability · Estimating probability

Give probabilities as fractions, decimals or percentages. ★ = challenge question.

1 Mia dropped a plastic cup 40 times and recorded how it landed.

Outcome	Frequency	Experimental probability
on its side	29	
upright	8	
upside down	3	

a Copy and complete the table.

b Mia predicted that the cup would be unlikely to land upright. Was she right? Explain.

c How could Mia make her estimates more reliable?

> **a** Experimental probability $= \dfrac{\text{frequency}}{\text{total}}$: side $\dfrac{29}{40} = 0.725$, upright $\dfrac{8}{40} = 0.2$, upside down $\dfrac{3}{40} = 0.075$.
>
> **b** Yes — upright happened in only 20% of drops, which is unlikely (less than $\frac{1}{2}$, and much less than landing on its side).
>
> **c** Repeat the experiment many more times — the more trials, the closer the estimate.

2 In an honesty experiment, 80 wallets were left in a street. 32 were handed in at a shop, 28 at a police station and 20 were kept.

a Work out the experimental probability of each outcome.

b Another wallet is left in the street. Estimate the probability that it is handed in somewhere.

> **a** Shop $\dfrac{32}{80} = 0.4$, police $\dfrac{28}{80} = 0.35$, kept $\dfrac{20}{80} = 0.25$ (they add to 1 ✓).
>
> **b** Handed in $= 0.4 + 0.35 = 0.75$.

3 In a video game a bonus item appears at random 3 times in every 24 levels. Work out the probability that the bonus appears in a particular level.

> $\dfrac{3}{24} = \dfrac{1}{8} = 0.125$.

4 In a survey, 8% of people who set themselves a fitness target achieved it. Another survey questions 350 people with a fitness target. How many would you expect to achieve it?

> Expected number = probability × number of trials $= 0.08 \times 350 = 28$ people.

5 A bakery bakes 500 loaves. 15 are burnt and thrown away, 380 are sold at full price and the rest are sold at a discount.

a Work out the relative frequency of each outcome.

b Estimate the probability that a loaf is *not* burnt.

c Next week the bakery bakes 2000 loaves. How many would you expect to be sold at a discount?

> **a** Discounted $= 500 - 15 - 380 = 105$. Burnt $\dfrac{15}{500} = 0.03$, full price $\dfrac{380}{500} = 0.76$, discounted $\dfrac{105}{500} = 0.21$.
> **b** $1 - 0.03 = 0.97$.
> **c** $0.21 \times 2000 = 420$ loaves.

6 The grades of 300 students in an exam were: A* 60, A 90, B 75, C 45, D 20, E 10.

a How many students got grade B or C?

b Estimate the probability that a student next year gets grade B or C.

c 150 students sit the exam next year. How many would you expect to get grade B or C?

> **a** $75 + 45 = 120$. **b** $\dfrac{120}{300} = 0.4$. **c** $0.4 \times 150 = 60$ students.

7 A dice is rolled 60 times and lands on 6 fifteen times.

a How many sixes would you expect from a fair dice in 60 rolls?

b What is the experimental probability of a 6?

c Does this prove the dice is biased? Explain.

> **a** Theoretical probability $\frac{1}{6}$: $\frac{1}{6} \times 60 = 10$.
> **b** $\dfrac{15}{60} = 0.25$ — higher than $\frac{1}{6} \approx 0.17$.
> **c** No. 60 rolls is a small number of trials, and a fair dice can easily give 15 sixes by chance. Roll it many more times (say 600); if the proportion stays near 0.25, that would be evidence of bias.

8 A spinner has three sectors of different sizes: red, blue and green. In 200 spins the results were red 96, blue 64, green 40.

a Estimate the probability of each colour.

b Which sector is the largest? Explain.

c How many greens would you expect in 1000 spins?

> **a** Red 0.48, blue 0.32, green 0.20.
> **b** Red — it came up most often, so it is the most likely outcome, which means the biggest sector (about half the spinner).
> **c** $0.2 \times 1000 = 200$.

Check: experimental probability = frequency ÷ total trials · all the probabilities of an experiment add up to 1 · expected number = probability × trials · more trials → better estimate.